Sub Code: 23EE/EC/CS2T02 **MIC23**

B. Tech II Semester Regular Examinations

CHEMISTRY

(EEE, ECE, CSE)

(Model Paper-I)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			CO	**BTL**	**Marks**
I.	**a)**	State Heisenberg uncertainty principle.	**CO1**	**L1**	**2 M**
	b)	On the basis of molecular orbital theory, explain that N_2 is diamagnetic.	**CO1**	**L2**	**2 M**
	c)	How supercapacitor is advantageous over a battery?	**CO2**	**L2**	**2 M**
	d)	What is meant by critical temperature?	**CO2**	**L1**	**2 M**
	e)	What is a potentiometric sensor? Give an example.	**CO3**	**L1**	**2 M**
	f)	Identify the technology used in polymer electrolyte membrane fuel cells (PEMFC). Give an example of a polymer electrolyte.	**CO3**	**L2**	**2 M**
	g)	Identify the catalyst used in coordination polymerization. Write its structure.	**CO4**	**L2**	**2 M**
	h)	List the applications of conducting polymers.	**CO4**	**L3**	**2 M**
	i)	Write the principle of UV-Visible spectroscopy.	**CO5**	**L2**	**2 M**
	j)	Write the selection rules in IR spectroscopy.	**CO5**	**L2**	**2 M**
		PART – B			**5 X 10 = 50 M**
		UNIT - I			
1.	**a)**	Derive Schrodinger wave equation of a microscopic particle in three-dimensional space.	**CO1**	**L3**	**5 M**
	b)	Describe the π-molecular orbitals of Butadiene. Identify its HOMO and LUMO.	**CO1**	**L3**	**5 M**
		(OR)			
2.	**a)**	Derive de Broglie's equation.	**CO1**	**L2**	**5 M**
	b)	Illustrate the energy level diagram of O_2. Calculate its bond order.	**CO1**	**L2**	**5 M**
		UNIT – II			
3.	**a)**	What is meant by doping? Write short notes on doped semiconductors.	**CO2**	**L2**	**5 M**
	b)	Outline the classification, properties, and applications of fullerenes.	**CO2**	**L2**	**5 M**
		(OR)			
4.	**a)**	What is a superconductor? List its applications.	**CO2**	**L4**	**5 M**
	b)	Discuss the classification of carbon nanotubes. List its applications	**CO2**	**L4**	**5 M**
		UNIT – III			
5.	**a)**	Find the concentration of Cu^{2+}solution in the given electrochemical cell: Mg/Mg^{2+} (0.01 M) // Cu^{2+}(M1)/Cu Given: $E^0_{Mg^{+2}/Mg} = -2.36V$; $E^0_{Cu^{+2}/Cu} = 0.34V$; $E_{cell} = 1.53V$ at 298K.	**CO3**	**L3**	**5 M**
	b)	Outline the construction and working of Zinc-air battery. List its uses.	**CO3**	**L4**	**5 M**
		(OR)			

6.	**a)**	What is an electrochemical sensor? Illustrate amperometric sensor with an example.	**CO3**	**L2**	**5 M**
	b)	What is a fuel cell? Explain the construction, working, and uses of H_2-O_2 fuel cell.	**CO3**	**L2**	**5 M**
UNIT – IV					
7.	**a)**	Discuss the preparation and properties of Bakelite. List its applications.	**CO4**	**L4**	**5 M**
	b)	What is a biodegradable polymer? Write short notes on polylactic acid.	**CO4**	**L2**	**5 M**
(OR)					
8.	**a)**	Describe the preparation and properties of Nylon-6,6.List itsapplications.	**CO4**	**L4**	**5 M**
	b)	Write short notes on polyhydroxyalkanoates.	**CO4**	**L2**	**5 M**
UNIT – V					
9.	**a)**	Outline the applications of UV-Visible spectroscopy.	**CO5**	**L2**	**5 M**
	b)	What is the principle involved in IR spectroscopy? List its applications.	**CO5**	**L4**	**5 M**
(OR)					
10.	**a)**	What is an electronic transition? Illustrate different types of electronic transitions in UV spectroscopy.	**CO5**	**L2**	**5 M**
	b)	Discuss the principle and classification of chromatography.List its applications.	**CO5**	**L4**	**5 M**

***** End of the Question Paper *****